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Investment Narratives and Cross-Market Risk Transmission: Evidence from the AI Sentiment Channel in Bitcoin and Technology Equities

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Statement of Originality

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Abstract

Investment narratives increasingly shape how investors position across asset classes, raising the question of whether a common narrative can weaken diversification by altering the way volatility is transmitted across otherwise distinct markets. We study this question using the emergence of artificial intelligence (AI) as a prominent investment narrative. Daily AI sentiment is constructed from Reddit posts and GDELT news headlines using a RoBERTa-based classifier and a dynamic factor model, and is then residualized on prediction-market expectations and macro-financial controls to proxy for its speculative component. Using data from April 2024 to March 2026, we apply ARMA-EGARCH models to test direct return and volatility effects, and TVP-VAR connectedness measures to examine volatility transmission between Bitcoin, the NASDAQ-100, and AI-exposed firms (NVIDIA, Alphabet, Microsoft). Expectation-adjusted AI sentiment significantly predicts next-day Bitcoin returns but not NASDAQ-100 returns or conditional volatility. More importantly, the association between sentiment and connectedness differs across the two systems: it is weakly negative in the Bitcoin–NASDAQ-100 system but positive and statistically significant in the targeted Bitcoin–AI-equity system. Within the targeted system, higher sentiment is associated with lower transmitted and higher received connectedness for Bitcoin, while transmitted connectedness from NVIDIA and Alphabet increases. These results indicate that a common investment narrative can reallocate rather than simply increase cross-market risk transmission, concentrating it among thematically exposed assets in ways that broad asset-class or index-level diversification measures fail to detect. We discuss the implications for dynamic diversification, thematic risk monitoring, and targeted stress testing in institutional portfolios, and present the AI case as a setting that motivates further research on narrative-driven risk under other dominant investment themes.

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1 Introduction

Financial markets are increasingly shaped by investment narratives that influence investor attention, expectations, and portfolio positioning across asset classes. Traditional portfolio theory emphasizes the diversification benefits of combining assets with imperfectly correlated return dynamics (Markowitz, 1952), but these benefits are not necessarily stable over time and may weaken when assets that appear distinct from a fundamental perspective become exposed to a common narrative or behavioral force. This possibility is particularly relevant for institutional investors, for whom the effectiveness of diversification depends critically on how cross-asset relationships evolve during periods of increased market attention and uncertainty. Existing research already documents return and volatility spillovers between Bitcoin, equity indices, and technology-related firms, suggesting that these markets are not fully independent (Bouri et al., 2018; Symitsi and Chalvatzis, 2018; Soltani and Boujelbène Abbes, 2026), while behavioral finance research shows that investor sentiment and attention can have particularly strong effects on assets characterized by high valuation uncertainty and limits to arbitrage (Baker and Wurgler, 2006; Akyildirim et al., 2024). Taken together, these findings raise a broader question for portfolio management: can a common investment narrative alter not only the risk of individual assets but also the structure and direction of risk transmission across otherwise distinct markets? The emergence of artificial intelligence (AI) as a major investment narrative provides a timely setting in which to examine this question. AI-related expectations have influenced valuations and investor attention toward technology firms, while cryptocurrencies, whose value is difficult to anchor in traditional fundamentals and is strongly influenced by expectations, narratives, and speculative demand, provide a distinct asset class through which to examine whether a common narrative can generate previously less visible cross-market risk links.

The analysis distinguishes between the general technology market and firms with direct exposure to the AI narrative. The NASDAQ-100 serves as a broad market representative, capturing technology-sector conditions rather than any specific narrative, while NVIDIA, Alphabet, and Microsoft are selected for their direct and well-documented exposure to AI. Comparing these two systems allows the analysis to separate narrative effects operating at the level of the general market from effects concentrated in individual, thematically exposed

firms.

However, important questions remain about how behavioral investment narratives shape cross-asset risk transmission. First, while previous studies have examined investor sentiment in cryptocurrency and equity markets, relatively little is known about whether a specific investment narrative can alter volatility connectedness across asset classes and, consequently, affect the diversification properties of multi-asset portfolios. This distinction is important because narrative-linked sentiment can be associated with common exposure among assets that are not conventionally classified as sharing the same fundamental risk. Second, sentiment measures can combine fundamentally driven expectations with more speculative or narrative-driven investor beliefs. Distinguishing these components is, therefore, important when assessing whether sentiment represents a broader market condition or an independent behavioral force that contributes to changes in cross-market risk transmission, a distinction particularly relevant for institutional investors seeking to identify hidden sources of common risk within diversified portfolios.

This study addresses these gaps by examining whether investment narratives can alter cross-market risk transmission and, in doing so, affect the diversification relationship between different asset classes. Using the emergence of AI as a prominent investment narrative as an empirical setting, we investigate whether expectation-adjusted AI sentiment is associated with changes in volatility connectedness between Bitcoin and technology-related equities. The analysis considers both a broad Bitcoin–NASDAQ-100 system and a more targeted system consisting of Bitcoin and selected AI-exposed technology firms, allowing us to determine whether narrative-driven sentiment affects cross-market risk broadly or whether its effects are concentrated among firms more directly exposed to the underlying investment theme. In this way, the study examines not only whether sentiment is associated with individual asset outcomes, but also whether it changes the structure and direction of risk transmission within a multi-asset investment environment.

The contribution of this study is threefold. First, it contributes to the behavioral finance literature by providing evidence that investor sentiment may influence financial markets not only through individual asset returns and volatility, but also through the structure and direction of volatility transmission across markets, a channel through which behavioral

forces can affect the diversification properties of multi-asset portfolios. Second, the study contributes to the literature on cross-asset connectedness by showing that the effects of a common investment narrative are not necessarily uniform across markets: distinguishing between broad technology-market exposure and firm-level exposure to a specific investment theme provides evidence that aggregate market indices may conceal important sources of cross-market risk. Third, the study develops an expectation-adjusted measure of AI sentiment by filtering raw sentiment through prediction-market information and macro-financial conditions, allowing the analysis to distinguish the speculative component of a prominent investment narrative from sentiment movements explained by broader economic conditions or observable expectations.

The findings have direct implications for institutional portfolio management and risk management. Diversification strategies often rely on historical correlations and average connectedness measures to assess whether different assets provide independent sources of risk, but if investment narratives can change the structure and direction of volatility transmission, these measures may provide an incomplete picture of portfolio exposure. The contrast between broad technology-market and firm-level connectedness we document indicates that aggregate indices may mask important sources of narrative-driven risk, implying that diversification should be evaluated dynamically and that thematic sentiment may be relevant for monitoring cross-asset exposures, stress testing, and risk allocation. While our empirical setting is necessarily anchored in a single episode, the post-ChatGPT rise of AI as an investment narrative, we treat AI as an illustrative case of a broader phenomenon rather than a self-contained subject of study, and expect the underlying mechanism to generalize to other episodes in which investor attention concentrates around a common theme, such as ESG investing, meme-stock speculation, or sector-specific technology cycles. The AI investment narrative therefore provides a setting to illustrate a broader challenge in modern portfolio management: identifying common behavioral exposures that are not captured by conventional asset-class classifications.

The remainder of the paper proceeds as follows. Section 2 reviews the relevant literature, Section 3 describes the data and sentiment construction, Section 4 presents the empirical methodology, Section 5 reports the results, and Section 6 discusses the implica-

tions, limitations, and conclusions.

2 Literature review

2.1 Sentiment, noise traders, and hard-to-value assets

A central pillar of traditional financial theory is the Efficient Market Hypothesis (EMH), which suggests that asset prices reflect available information through the trading activity of market participants (Fama, 1970). However, the persistence of market anomalies and speculative bubbles prompted the rise of behavioral finance. A crucial development in this field is the Noisy Trader Framework (De Long et al., 1990), where markets are populated by both rational arbitrageurs and noise traders whose demand is driven by sentiment. In this model, sentiment, defined as the broad emotional outlook of participants, acts as a noisy signal that introduces additional volatility, suggesting a direct link between emotions and volatility.

Building on this, the work of Baker and Wurgler (2006) established investor sentiment as an important determinant of asset prices. Importantly, sentiment does not uniformly impact the entire market; rather, its impact is most pronounced on assets that are difficult to value or costly to arbitrage (Baker and Wurgler, 2006). Given that both technology stocks and cryptocurrencies are frequently characterized by high levels of innovation and uncertainty, they are likely to be sensitive to investor sentiment.

2.2 Sentiment in cryptocurrency and technology equity markets

The empirical evidence supporting these theoretical frameworks is substantial, though the practical measurement of investor sentiment remains a complex challenge. As Guler (2021) points out, there is no single proxy that fully captures investor sentiment. Consequently, the literature has adopted various proxies, ranging from traditional indicators such as trading volume and surveys to more modern machine learning techniques. Within the cryptocurrency market, these proxies have consistently demonstrated that sentiment is significantly associated with both returns and risk. For example, Guler (2021) utilized the American Association of Individual Investors (AAII) sentiment alongside the Fear & Greed

Index and the trading volume to show a positive relationship between sentiment and daily Bitcoin returns between 2018 and 2020. This research used an EGARCH(1,1) specification to capture asymmetric volatility effects, finding that pessimistic shifts in the market mood not only lower prices but also affect the asset’s risk profile more than equivalent optimistic shifts.

Moreover, the role of sentiment as a determinant of Bitcoin returns and volatility is not limited to individual cryptocurrencies. Akyildirim et al. (2024) utilize a TVP-VAR connectedness framework to capture dynamic spillovers between major cryptocurrencies. They regress directional spillovers on sentiment derived from Thomson Reuters MarketPsych Indices (TRMIs), showing that sentiment is significantly associated with return and volatility spillovers within the cryptocurrency market. Long et al. (2024) take a similar approach but include investor sentiment, derived from Reddit posts in combination with VADER and TextBlob, as an endogenous variable in the connectedness system with five cryptocurrencies. This approach analyses the dynamic role of investor sentiment, and researchers find that sentiment is a recipient of volatility rather than a transmitter. A similar pattern emerges when examining tech-heavy equities and broader market indices. Research by Deveikyte et al. (2022) found that sentiment, extracted from daily news headlines and social media, was significantly correlated with both the volatility and daily returns of the FTSE-100 in 2019. Thus, the literature suggests that sentiment is relevant in both cryptocurrency and equity markets, providing a basis for studying whether sentiment may also contribute to cross-market connectedness.

2.3 Connectedness between cryptocurrencies and technology equities

Building on this relationship, research has documented stronger links between digital assets and technology equities, often interpreting their relationship as part of a broader risk-on complex (Bouri et al., 2018). This connection may reflect both financial-market co-movement and economic links. For example, Symitsi and Chalvatzis (2018) identify bidirectional return and volatility spillovers between Bitcoin and hardware-related technology stocks, partially attributing this relationship to demand for semiconductors from cryp-

tocurrency mining. Similarly, Uzonwanne (2021) documents significant daily volatility and return spillovers between the S&P 500 and Bitcoin. More recent TVP-VAR evidence from Akyildirim et al. (2025) shows that connectedness between cryptocurrencies and NASDAQ Blockchain Economy Index stocks increased substantially during the COVID-19 pandemic, with volatility spillovers running mainly from cryptocurrencies to equities. However, other studies view cryptocurrencies and equities as relatively distinct asset classes (Wang et al., 2019; Baur et al., 2018). Thus, the strength and direction of the connection between these markets remain an open empirical question.

2.4 AI narratives and the post-ChatGPT period

The differing results of the aforementioned studies may reflect differences in sampling periods. The release of ChatGPT in late 2022 accelerated public and investor attention towards generative AI. This shift is relevant because AI-related assets are often difficult to value, due to uncertainty about their long-term relevance and profitability (Klass et al., 2025). Following Baker and Wurgler (2006), such assets may be particularly susceptible to investor sentiment; hence, the rise of AI could strengthen the link between the two asset classes.

In support of this, Soltani and Boujelbène Abbes (2026) use a generalized VAR-based spillover framework for the NASDAQ-100, S&P 500, and major cryptocurrencies and show that total market connectedness reached a peak in early 2023, coinciding with the rapid expansion of generative AI. The study further incorporates Russian investor sentiment as an endogenous variable and finds that sentiment is a net transmitter of shocks. This suggests that investor sentiment may be an important channel for volatility spillovers in the post-ChatGPT period. Similarly, Wang et al. (2025) use a GJR-GARCH model, to show that AI-related incidents had no significant impact on the returns of cryptocurrencies before late 2022, but began to trigger significant negative return shocks after the release of ChatGPT. To some extent, the relationship between stock returns and AI attention may depend on the link between AI and a firm’s business model (Junanta and Rokhim, 2025). Taken together, these results suggest that the tech-crypto link identified in earlier literature may also involve a sentiment-based transmission channel that connects technology equities and cryptocurrencies through their shared exposure to AI.

2.5 Narrative-driven risk beyond AI

The AI case studied here is best understood as one instance of a broader pattern documented in the narrative-economics and behavioral-finance literature. Shiller (2017, 2019) argues that widely shared narratives can act as an independent economic force, spreading through investor populations much like epidemics and generating waves of speculative activity that are not fully explained by fundamentals. Consistent with this view, empirical work on other thematic episodes has documented similar patterns of narrative-driven co-movement: meme-stock episodes have been shown to generate correlated retail-driven volatility across otherwise unrelated names, and ESG-labeled assets have exhibited periods of elevated co-movement tied to shifts in sustainability-related sentiment and fund flows rather than to shared fundamentals. These findings suggest that the mechanism examined in this paper (a common narrative altering the structure of cross-market risk transmission rather than uniformly raising it) is not specific to AI or cryptocurrency markets, but reflects a more general channel through which investor attention can create hidden, time-varying sources of portfolio concentration.

2.6 Isolating speculative sentiment

While the literature establishes AI sentiment as a potential catalyst, it often treats sentiment as a standalone aggregate force. However, following the Noisy Trader Framework, it is important to isolate the part of sentiment that is not explained by market conditions. This residual component can be interpreted as a proxy for irrational sentiment. Existing studies commonly distinguish between rational and irrational sentiment through orthogonalization using market variables, finding that both components may influence Bitcoin volatility (Guler, 2021). More recently, Akyildirim et al. (2024) use source-based proxies, suggesting that news-derived sentiment may contain a more informational signal, while social-media sentiment may capture more speculative noise. Yet, these proxies are limited: news sentiment is often correlated with social media trends (Akyildirim et al., 2024), and econometric filters remain sensitive to model specification and omitted variables.

To address these limitations, prediction-market information offers a promising mechanism for filtering sentiment. Unlike raw social-media sentiment, prediction markets attach

financial incentives to beliefs about future outcomes. Wolfers and Zitzewitz (2004) argue that such incentives can improve the aggregation of dispersed information.

2.7 Transmission mechanism

Based on the literature discussed above, this study views speculative AI sentiment as a potential channel through which technological narratives may affect cross-market volatility transmission. The proposed mechanism is that AI-related sentiment increases investor attention to AI topics and AI-exposed assets. This attention may then affect trading behavior, as investors adjust positions in response to perceived opportunities, uncertainty, or changing beliefs about future AI-related growth. When similar AI-related narratives influence both cryptocurrency markets and AI-exposed technology equities, volatility shocks in these markets may become more synchronized. This can result in stronger cross-market spillovers, while the resulting market movements may also feed back into sentiment.

Positive sentiment toward AI as a technology does not necessarily imply positive investor sentiment toward AI-exposed equities. A text about capabilities, risks or societal implications can have a different sentiment orientation from its implications for stock valuations. An attention channel offers one possible interpretation, consistent with the behavioral motivation in De Long et al. (1990): AI-related discussion may draw attention to exposed assets and affect trading behavior. The signed sentiment index does not directly measure attention volume or investor positions, however, so the conceptual transmission mechanism remains a hypothesis rather than an identified causal sequence.

This conceptual channel is summarized in Figure 1.

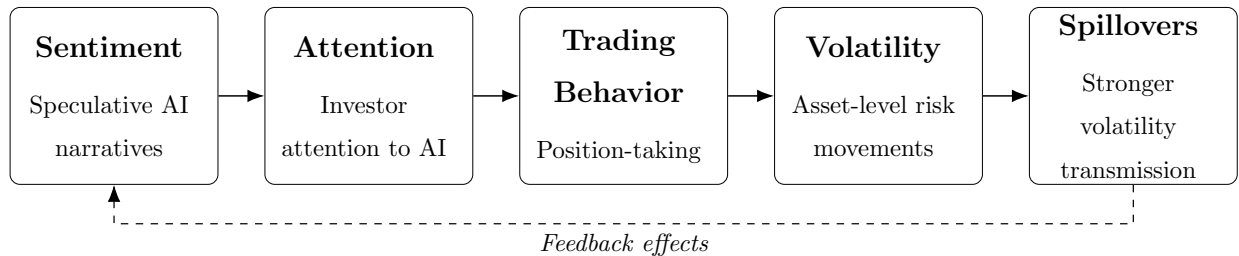


Figure 1: Conceptual Transmission Mechanism of AI Sentiment on Cross-Market Spillovers

2.8 Research gap and empirical focus

Taken together, the literature suggests that sentiment, volatility, and connectedness are closely related, but it leaves an important distinction unresolved. Existing sentiment studies show that investor mood matters for cryptocurrency and equity markets, especially for assets that are difficult to value or costly to arbitrage. Existing connectedness studies further show that Bitcoin and technology-related equities may transmit shocks to one another, although the strength and direction of this relationship differ across periods and asset groups. What remains less clear is whether AI-specific sentiment represents a common narrative channel between Bitcoin and technology equities, or whether it mainly reflects broader market optimism already captured by observable expectations and macro-financial conditions.

This study therefore examines whether expectation-adjusted AI sentiment is associated with return-volatility dynamics and volatility connectedness in two systems: a broad Bitcoin–NASDAQ-100 system and a more targeted Bitcoin–AI-equity system. Comparing these systems allows the analysis to assess whether AI sentiment is mainly related to general technology-market exposure or to firms more directly linked to the AI narrative.

3 Data

This study uses a daily data set that combines textual sentiment data, financial asset prices, prediction-market variables, and macro-financial controls. The main sample covers the period from April 1, 2024, to March 31, 2026. Because Kalshi data are only available from April 18, 2024, specifications using prediction-market information are estimated over a slightly shorter sample. Asset prices and market controls are obtained from Yahoo Finance, the Geopolitical Risk (GPR) and Economic Policy Uncertainty (EPU) indices are obtained from their official repositories, textual data are collected from Reddit and GDELT, and prediction-market data are sourced from Kalshi and Metaculus.

All variables are aligned with the US equity-market trading calendar. Continuous series such as Bitcoin prices, GPR, EPU, and prediction-market variables are restricted to US trading days. Textual observations are assigned to US equity trading days before source-

specific sentiment averages are calculated, as detailed in Section 3.1.

3.1 Data for the Sentiment Index

Textual data are collected from Reddit and GDELT using the keyword filter reported in Table A1. Reddit data consist of concatenated titles and body texts from the r/technology, r/artificial, and r/ArtificialIntelligence subreddits, retrieved through the Pushshift API. News data consist of English-language GDELT headlines obtained through Google BigQuery, with a daily cap of 100 headlines. After removing duplicates, the final textual dataset contains 58,014 news headlines and 34,393 Reddit posts.

Because US equity markets are closed on weekends and public holidays, textual observations collected on non-trading days are pooled with the following trading day’s observations. Each text is scored individually and assigned to the next available equity trading date, including its own date when it is a trading day; scores are then averaged across all assigned texts within each source. This rolling-forward procedure retains the in-sample textual observations while synchronizing the sentiment series with the daily asset-return and volatility data. Weekend and holiday texts therefore receive the same observation-level weight as texts collected on the following trading day.

AI sentiment is not directly observed and is therefore constructed from the textual information in Reddit posts and GDELT news headlines. Sentiment classification is performed using the Cardiff Twitter-RoBERTa transformer model, based on the BERT and RoBERTa architectures of Devlin et al. (2018) and Liu et al. (2019). For each text observation, the model returns probabilities for positive, neutral, and negative sentiment. The observation-level score is the positive probability minus the negative probability, so higher values indicate more positive AI-related sentiment. The formal definition is reported in Appendix A.1.

To assess whether the RoBERTa classifier captures the intended AI-related sentiment, a manual validation is conducted on a sample of 100 observations, consisting of 50 GDELT headlines and 50 Reddit posts. Each observation is manually labeled as positive, neutral, or negative based on its tone toward AI. In addition, the five most positive and five most negative observations per source are inspected to assess whether extreme RoBERTa scores

correspond to sentiment signals.

The validation results are reported in Table A2, which shows that the agreement between the discrete RoBERTa label and the manual label equals 72%. The classifier assigns a neutral label more often than the manual review, which suggests that it is relatively conservative in classifying texts as clearly positive or negative. Importantly, polarity reversals, defined as manually positive texts classified as negative or manually negative texts classified as positive, occur in 2 out of 100 observations. Since the sentiment index uses this continuous score rather than only the discrete model label, the validation exercise supports the use of RoBERTa scores as a noisy but directionally meaningful measure of AI-related sentiment.

The manual validation is based on 100 observations and should be interpreted as an indicative, rather than definitive, check on classifier performance. The 72% agreement rate and the small number of polarity reversals support its use as a noisy directional measure, but a larger independently labeled sample would permit a more precise estimate of classification accuracy.

The inspection of extreme observations provides additional validity. Among the 20 most extreme observations, consisting of the five most positive and five most negative texts from each source, the model label agrees with the manual label in 18 cases and there are no polarity reversals. Some Reddit extremes also contain promotional or toxic language, confirming that individual text-level scores can be noisy.

For each text observation, the model returns probabilities for positive, neutral, and negative sentiment. The observation-level sentiment score is defined as the positive probability minus the negative probability, so that higher values indicate more positive AI-related sentiment. The formal observation-level definition and daily source-aggregation rule are reported in Appendix A.1.

Rather than relying on either Reddit or GDELT in isolation, this study combines both series to extract their common component of AI-related sentiment and to reduce the influence of source-specific noise. To accommodate missing observations and the autocorrelation present in the daily sentiment measures, a one-factor Dynamic Factor Model is estimated using the Expectation-Maximization approach of Banbura and Modugno (2014).

The model consists of a measurement equation linking the source-specific scores to a latent common factor and an autoregressive state equation for that factor. The complete specification is reported in Appendix A.2.

The lag order of the state equation is selected using BIC. Missing observations are handled within the Kalman-filtering and EM-estimation procedure, and the sign of the latent factor is normalized so that higher values correspond to more positive sentiment. The resulting sign-corrected factor is used as the AI Sentiment Index, AIS_t .

In addition to the raw sentiment index, this study constructs an expectation-adjusted AI sentiment measure. This measure is obtained by residualizing the AIS on prediction-market expectation variables and macro-financial controls:

$$AIS_t = \alpha + \beta_1 Kalshi_t + \beta_2 Metaculus_t + \gamma' X_t + \varepsilon_t, \quad (1)$$

where the prediction-market controls $Kalshi_t$ and $Metaculus_t$ are the first difference of the Kalshi before-2030 probability and the first difference of the inverted Metaculus expected timing measure. The macro-financial controls are S&P 500 returns, DXY returns, VIX, $\ln(EPU)$, and $\ln(GPR)$. This residual is used as a macro- and expectation-adjusted measure of AI sentiment (EAIS). Following the filtering logic of Baker and Wurgler (2006), it removes the variation explained by the included controls, but does not by itself identify speculative or irrational beliefs.

Table 1 reports descriptive statistics and diagnostics for the raw sentiment series and the constructed indices. The Reddit series is complete over the trading-day sample, while the GDELT series contains 10 missing observations due to a database outage in mid-2025. The presence of missing observations and strong serial correlation in the raw series motivates the use of a dynamic factor model estimated by the EM algorithm. The resulting AI Sentiment Index is standardized to have zero mean and unit variance. For the one-factor dynamic factor model, the lag length is selected using BIC, which selects $p = 1$. The selected model produces positive factor loadings for both the GDELT news sentiment and the Reddit sentiment, indicating that the extracted factor captures their common AI-related sentiment component. The Reddit loading is larger than the GDELT loading, suggesting that the baseline AIS is more strongly driven by social-media sentiment. Full factor-selection results are reported in Table A4.

Table 1: Descriptive Statistics and Diagnostics of Sentiment Variables

Statistic	Raw		Constructed Indices	
	GDELT	Reddit	AIS	EAIS
N	492	502	502	488
Missing Days	10	0	0	0
Mean	0.101	0.040	0.000	0.000
Std. Deviation	0.083	0.112	1.000	0.684
Minimum	-0.177	-0.303	-1.902	-3.591
Maximum	0.347	0.396	2.053	1.923
Skewness	-0.383	0.199	0.165	-0.209
Kurtosis	3.234	2.766	1.839	4.083
Jarque-Bera	12.91**	4.53	30.49***	26.37***
Ljung-Box $Q(5)$	250.96***	931.05***	2288.90***	1125.32***
Ljung-Box $Q(10)$	379.71***	1778.31***	4329.32***	1955.72***
Ljung-Box $Q(20)$	516.10***	3332.67***	7871.62***	3303.81***
ADF Statistic	—	—	-1.438	-4.094***
Correlation with AIS	0.482	0.824	1.000	0.712

Notes: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table A3 shows that the prediction-market variables, DXY and S&P 500 returns are individually insignificant, while GPR, VIX and EPU are significant at conventional levels. The adjustment therefore operates primarily through macro-financial, geopolitical and policy-uncertainty conditions, rather than through a separately demonstrated prediction-market contribution. EAIS is interpreted accordingly as a macro- and expectation-adjusted measure. One possible explanation for the insignificant prediction-market coefficients is that AGI-timing expectations contain relatively little information about day-to-day fluctuations in textual tone. EAIS remains the main sentiment measure and AIS is retained for the robustness checks.

Table 1 shows that the EAIS remains positively correlated with the AIS, while displaying lower serial correlation and stronger evidence of stationarity. Although the ADF test fails to reject a unit root for the AIS, both sentiment indices are retained in levels to maintain

comparability as AIS is only used as a robustness check.

3.2 Financial Data

Financial data consist of Bitcoin, the NASDAQ-100, and three AI-exposed NASDAQ-100 companies: NVIDIA, Alphabet, and Microsoft. Daily adjusted closing prices are transformed into log returns, $r_t = \ln(P_t/P_{t-1})$. The return transformation reduces the full financial sample to 501 observations, while specifications using expectation-adjusted sentiment are estimated over 487 observations due to prediction-market availability. Table 2 reports descriptive statistics and diagnostics for both returns and conditional volatilities, constructed using univariate EGARCH(1,1) specifications.

The same-size comparison adds Cisco Systems (CSCO), Booking Holdings (BKNG) and Mondelez International (MDLZ), using Yahoo Finance adjusted prices and the identical trading calendar. These firms span networking, travel services and packaged foods and have substantially fewer mentions in this study’s AI-filtered text corpus than NVIDIA, Alphabet and Microsoft. They provide a comparison basket rather than a market-capitalization- or sector-matched control group; Cisco also has AI-infrastructure exposure. Section 5.4.5 reports the selection check and comparison results.

Table 2: Descriptive Statistics and Diagnostics of Asset Returns and Conditional Volatilities

Statistic	Bitcoin (BTC)		NASDAQ-100 (NDX)		NVIDIA (NVDA)		Alphabet (GOOGL)		Microsoft (MSFT)	
	Ret	Vol	Ret	Vol	Ret	Vol	Ret	Vol	Ret	Vol
Mean	-0.0000	0.0295	0.0005	0.0124	0.0013	0.0298	0.0012	0.0188	-0.0002	0.0155
Std. Dev.	0.0304	0.0052	0.0134	0.0041	0.0312	0.0080	0.0188	0.0026	0.0153	0.0030
Skewness	0.1300	1.2647	0.5142	1.3708	-0.2980	0.8578	0.3236	0.8297	-0.3748	-0.2406
Kurtosis	6.4700	6.5685	14.3327	5.9810	7.6374	3.0968	6.9095	3.2709	12.0694	2.5630
Jarque-Bera	246.06***	391.56***	2644.25***	336.33***	445.13***	61.21***	319.51***	58.51***	1690.21***	8.93**
ADF Statistic	-23.99***	-5.24***	-24.27***	-4.41***	-14.94***	-2.95**	-22.28***	-4.31***	-21.91***	-1.39
KPSS Statistic	0.235	0.252	0.067	0.120	0.072	1.002***	0.145	0.275	0.219	0.401*
Ljung-Box $Q(5)$	3.50	1353.98***	16.35***	1566.66***	21.36***	2053.49***	4.05	1619.05***	1.07	2203.15***
Ljung-Box $Q(10)$	10.86	1836.66***	19.32**	2273.29***	24.04***	3444.56***	6.32	2513.77***	1.74	4031.56***
Ljung-Box $Q(20)$	31.46**	2024.56***	31.69**	2686.55***	34.02**	5309.57***	22.90	3556.93***	17.73	6844.54***
ARCH-LM(5)	24.58***	—	53.50***	—	16.27***	—	2.67	—	10.68*	—
ARCH-LM(10)	25.28***	—	58.35***	—	16.88*	—	5.35	—	12.39	—
ARCH-LM(20)	30.25*	—	59.63***	—	20.01	—	14.15	—	29.22*	—

Notes: $N = 501$ for all series. ‘Ret’ denotes log returns and ‘Vol’ denotes conditional volatility. Dashes (—) indicate the ARCH-LM test is not applicable to conditional volatility series. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

The return series display the standard features of financial time series. Jarque-Bera tests reject normality for all assets, and the return distributions are characterized by substantial excess kurtosis. This supports the use of Student’s t innovations in the ARMA-EGARCH specifications. The ADF tests indicate that the log return series are stationary, while the ARCH-LM tests provide evidence of conditional heteroskedasticity and volatility clustering. These diagnostics motivate the use of EGARCH models for the univariate return-volatility analysis.

The volatility series are used as inputs for the TVP-VAR connectedness framework. Post-estimation diagnostics for the EGARCH specifications are reported in Table A5 and generally indicate limited remaining serial dependence or ARCH effects in the standardized residuals.

3.3 Control Variables

The empirical analysis includes macro-financial controls to account for broad market conditions and aggregate uncertainty. S&P 500 returns are included to capture general equity-market movements, DXY returns capture changes in the US dollar, the VIX captures market-wide risk aversion, and the logarithms of GPR and EPU capture geopolitical and policy uncertainty. The S&P 500 and DXY are transformed into log returns, GPR and EPU enter in logarithms, and the VIX is kept in levels.

Forward-looking AI expectations are measured using the Kalshi contract price and Metaculus community forecasts. The Kalshi variable is the price, in cents, of the “Before 2030” contract on OpenAI announcing that it has attained AGI. The price is interpretable as a market-implied probability in percentage points, subject to trading frictions. The Metaculus variable uses the recency-weighted median forecast for the date on which the first general AI system will be devised, tested and publicly announced. We measure the number of days from each forecast timestamp to its predicted date and change the sign, so that higher values indicate nearer-term expected developments; “inverted” means negative days, rather than a reciprocal. The last daily forecast is retained and both expectation series are carried forward before alignment with equity trading dates. Both series are first-differenced because their levels are highly persistent. Kalshi aggregates financially backed

positions, whereas this Metaculus series aggregates community forecasts without a traded contract price. They concern different events and are included to capture expectation changes from these distinct sources. Descriptive statistics and diagnostic tests are reported in Table A6, which shows mixed evidence of stationarity for $\ln(EPU)$.

4 Methodology

This section outlines the econometric models and methodological procedures used in this study. The empirical analysis proceeds in two steps. First, univariate ARMA-EGARCH models are estimated to examine return and volatility dynamics. Second, a time-varying parameter vector autoregressive model (TVP-VAR) is used to analyze dynamic spillovers.

4.1 ARMA-EGARCH model

To examine how AI-related sentiment affects the returns and volatility of Bitcoin and the NASDAQ-100, which is useful because sentiment may affect individual markets without changing volatility transmissions, this study uses an ARMA-EGARCH model. ARCH/GARCH models are well suited to financial time series with heteroskedasticity and volatility clustering (Engle, 1982; Bollerslev, 1986), while the EGARCH specification allows positive and negative shocks to affect volatility asymmetrically and ensures positive conditional variance without imposing non-negativity restrictions (Nelson, 1991). The mean equation is specified as

$$r_t = \mu + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \lambda S_{t-1} + \varepsilon_t, \quad (2)$$

where r_t denotes the log return of the asset at time t , and S_{t-1} denotes the lagged AI sentiment variable. Lagged sentiment is used to preserve the temporal ordering between sentiment and financial-market outcomes and to reduce endogeneity concerns. In the benchmark specification, S_{t-1} is the expectation-adjusted AI Sentiment Index, while the raw AI Sentiment Index is used as a robustness check. The coefficient λ measures whether AI sentiment is associated with next-day returns. Economically, a positive estimate implies that optimistic AI-related sentiment predicts higher subsequent returns, whereas a neg-

ative estimate suggests weaker subsequent returns, potentially reflecting overreaction or subsequent corrections. The error term is defined as

$$\varepsilon_t = \sigma_t z_t, \quad (3)$$

where z_t follows a standardized Student's t -distribution to account for fat tails in financial returns. The EGARCH(1,1) variance equation is

$$\ln(\sigma_t^2) = \omega + \beta \ln(\sigma_{t-1}^2) + \alpha \left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - E|z_{t-1}| \right) + \gamma \frac{\varepsilon_{t-1}}{\sigma_{t-1}} + \eta S_{t-1} \quad (4)$$

Here, β captures volatility persistence, α captures the effect of shock magnitude, and γ measures asymmetry. A negative γ is consistent with the traditional leverage effect. The coefficient η measures whether AI sentiment directly affects conditional volatility. Including S_{t-1} in both the mean and variance equations allows the analysis to distinguish between return effects and volatility effects.

The ARMA lag orders are selected using BIC. The robustness specifications use the same lag order as the benchmark model to ensure comparability. The EGARCH order is fixed at EGARCH(1,1), a standard specification in financial volatility applications. The next subsection presents the time-varying parameter VAR framework used to investigate cross-market dynamics.

4.2 Time-varying parameter VAR model and connectedness measures

The second part of the empirical analysis studies dynamic volatility spillovers between Bitcoin and technology equities using a TVP-VAR model (Primiceri, 2005). In economic terms, these spillovers measure the extent to which volatility shocks in one market contribute to volatility in another. Higher connectedness implies stronger cross-market risk transmission and weaker diversification benefits. A time-varying framework is appropriate because the relevance of AI-related information for technology equities and Bitcoin may change over time.

The methodology follows Koop and Korobilis (2014) and the connectedness framework of Diebold and Yilmaz (2009, 2012), as extended by Antonakakis et al. (2020). A simpler

static VAR would impose constant spillover relationships, while a rolling-window approach would require an arbitrary window length. The TVP-VAR is better suited to the post-ChatGPT setting because the importance of AI-related information is likely to evolve over time.

Depending on the specification, the vector of endogenous variables contains either the conditional volatilities of Bitcoin and the NASDAQ-100, or the conditional volatilities of Bitcoin and the selected AI-exposed NASDAQ-100 firms: Microsoft, NVIDIA, and Alphabet. The lag length is selected using BIC. Time variation in the coefficients is modeled as a random walk, and the model is estimated using Kalman filtering and forgetting factors following Koop and Korobilis (2014). The complete observation and state equations are reported in Appendix A.3.

Connectedness is obtained from the time-varying vector moving-average representation of the TVP-VAR. Generalized forecast error variance decompositions are computed following Pesaran and Shin (1998). This approach is invariant to variable ordering and therefore does not require a recursive identification structure. Because the resulting variance-decomposition rows do not necessarily sum to one, they are normalized before the connectedness measures are calculated.

The analysis considers total connectedness, directional connectedness transmitted to other variables, directional connectedness received from other variables, net connectedness, and net pairwise directional connectedness. Total connectedness measures the share of forecast-error variance attributable to cross-variable shocks. The directional measures distinguish shock transmission from shock reception, while the net measures identify the dominant direction of volatility transmission. The complete definitions are reported in Appendix A.4.

The benchmark analysis sets the forecast horizon to $H = 10$, following Akyildirim et al. (2024). As a robustness check, the analysis is repeated using $H = 100$, following Long et al. (2024).

The TVP-VAR is estimated using conditional volatility series obtained from EGARCH(1,1) models, as described in the Data section. These volatility inputs do not include sentiment regressors, ensuring that the connectedness measures are not mechanically driven by

sentiment in the volatility model. Following Akyildirim et al. (2024), the analysis then examines whether AI-related sentiment explains variation in volatility connectedness using the regression equation:

$$Spillover_{i,t} = \alpha + \beta_1 Sent_t + \beta_2 Sent_{t-1} + \gamma' X_t + u_t, \quad (5)$$

where $Spillover_{i,t}$ denotes the relevant connectedness measure. The main analysis focuses on TCI_t , $TO_{i,t}$, and $FROM_{i,t}$, which distinguish between overall connectedness, shock transmission, and shock reception. The vector X_t contains macro-financial controls that may affect connectedness between technology assets and cryptocurrency markets: DXY returns, $\ln(GPR)$, $\ln(EPU)$, and VIX.

S&P 500 returns enter the orthogonalization regression (Equation (1)) to remove linear co-movement between raw AI sentiment and the broad equity market at construction. The connectedness regression (Equation (5)) serves a different purpose: it relates the adjusted sentiment measure to spillovers, conditional on the existing set of dollar and uncertainty controls. VIX captures broad market uncertainty in this second equation, whereas S&P 500 returns capture realized equity-market movements. Their inclusion in different stages reflects this specification choice.

The coefficient β_1 (β_2) measures whether (lagged) AI sentiment is associated with stronger or weaker volatility spillovers, conditional on these controls. A positive coefficient suggests that AI sentiment amplifies cross-market risk transmission, whereas a negative coefficient indicates more segmented or asset-specific volatility dynamics. The baseline specification uses both contemporaneous and lagged expectation-adjusted sentiment, while robustness specifications use the raw AI Sentiment Index.

5 Empirical results

This section presents the empirical results. It first examines the direct return and volatility effects of expectation-adjusted AI sentiment using ARMA-EGARCH models. It then analyzes time-varying volatility connectedness between Bitcoin and technology equities with TVP-VAR connectedness measures, before regressing these measures on AI sentiment and controls. Finally, robustness checks are conducted.

5.1 Direct effects: ARMA-EGARCH results

This subsection examines whether AI-related sentiment is directly associated with the return and volatility dynamics of Bitcoin and the NASDAQ-100. As a preliminary diagnostic, the one-day Granger causality tests in Table A7 indicate that expectation-adjusted AIS contains predictive information for Bitcoin returns, but not for NASDAQ-100 returns. This supports the use of lagged expectation-adjusted sentiment for Bitcoin, and the same specification is also used for the NASDAQ-100 to maintain comparability.

For each asset, the ARMA order in the conditional mean equation is selected using BIC, while the conditional variance is modeled using an EGARCH(1,1) specification with Student's t innovations. The BIC selects an ARMAX(0,1)-EGARCHX(1,1) model for Bitcoin and an ARMAX(0,0)-EGARCHX(1,1) model for the NASDAQ-100.

Table 3 shows that expectation-adjusted AI sentiment has a positive and statistically significant effect on Bitcoin returns, with $\lambda = 0.427$. Because returns are multiplied by 100 in the ARMA-EGARCH estimations, this implies that a one-unit increase in EAIS is associated with a 42.7 basis points increase in next-day Bitcoin returns, consistent with the sentiment-return channel documented by Guler (2021). In contrast, the corresponding coefficient for the NASDAQ-100 is positive but statistically insignificant. This suggests that unexpected AI sentiment is not simply capturing general optimism toward technology stocks, but is more strongly reflected in the pricing of Bitcoin.

The variance equation provides no evidence that AI sentiment directly affects log conditional variance. The sentiment coefficient in the variance equation is positive but statistically insignificant for both Bitcoin, $\eta = 0.027$, and the NASDAQ-100, $\eta = 0.013$. Thus, while expectation-adjusted AI sentiment predicts next-day Bitcoin returns, it does not provide statistically significant evidence of a direct variance effect in either asset within the EGARCH specification.

Table 3: ARMA-EGARCH Estimation Results EAIS

	Bitcoin	NASDAQ-100
μ	-0.016 (0.105)	0.127*** (0.043)
θ_1	-0.075** (0.036)	— —
λ	0.427*** (0.162)	0.083 (0.066)
ω	0.195 (0.206)	0.002 (0.014)
α	-0.150* (0.081)	-0.236*** (0.035)
β	0.907*** (0.095)	0.944*** (0.019)
γ	0.036 (0.055)	0.028 (0.039)
η	0.027 (0.030)	0.013 (0.019)
ν	5.147*** (1.145)	5.189*** (1.401)
N	486	486
Log-likelihood	-1180.110	-743.618
BIC	4.971	3.162
$Q_z(10)$	8.466	6.392
$Q_z^2(10)$	13.378	11.322
$Q_z(20)$	32.315**	12.731
$Q_z^2(20)$	17.330	14.314

Notes: Robust standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. $Q_z(k)$ and $Q_z^2(k)$ denote Ljung-Box statistics for standardized and squared standardized residuals at lag k .

The remaining parameter estimates are consistent with common features of financial return series. Volatility is highly persistent for both Bitcoin and the NASDAQ-100, with $\beta = 0.907$ and $\beta = 0.944$, respectively. These estimates imply that volatility shocks are not absorbed immediately, but decay gradually over time. The implied half-life is approximately

seven trading days for Bitcoin and twelve trading days for the NASDAQ-100, indicating that volatility shocks persist longer in the technology equity index than in Bitcoin during the sample period. The asymmetric volatility parameter is not statistically significant in either model, providing no evidence of leverage effects. Finally, the estimated degrees-of-freedom parameters are close to five for both assets, indicating substantial tail risk and supporting the use of Student’s t innovations.

Post-estimation diagnostics generally support the selected models. For the NASDAQ-100, the Ljung-Box tests on standardized and squared standardized residuals do not indicate remaining serial dependence. For Bitcoin, the squared residual tests show no remaining ARCH effects, although the Ljung-Box test on standardized residuals at 20 lags is significant at the 5% level. The Bitcoin results should therefore be interpreted with some caution, but the volatility specification appears adequate.

5.2 Connectedness dynamics: TVP-VAR results

This subsection examines time-varying volatility connectedness between Bitcoin and technology equities. Two systems are considered: a BTC–NASDAQ-100 benchmark system and a BTC–AI-equity system consisting of Bitcoin, NVIDIA, Alphabet, and Microsoft. The NASDAQ-100 serves as a broad, technology-heavy equity benchmark, while NVIDIA, Alphabet, and Microsoft represent more targeted exposure to the AI trade. For both systems, BIC selects a TVP-VAR(1) specification, and connectedness measures are computed using a forecast horizon of $H = 10$.

Table 4 shows that the BTC–NDX benchmark system has an average TCI of 18.79%. In this system, the NASDAQ-100 is the average net transmitter of volatility, while Bitcoin is the net receiver. Connectivity is stronger in the BTC–AI-equity system, where the TCI is 27.43%, which means that 27.43% of the 10-day-ahead forecast-error variance is explained by cross-market volatility. This suggests that the broad technology index masks stronger firm-level volatility links between Bitcoin and AI-exposed equities.

To address the possibility that the higher connectedness in the BTC–AI-equity system reflects its larger number of variables, Section 5.4.5 reports a comparison system of identical size using Bitcoin, Cisco, Booking and Mondelez. Panel C reports its average connectedness

alongside the two original systems.

Table 4: Average TVP-VAR Connectedness Table, $H = 10$

<i>Panel A: BTC-NDX system</i>					
	BTC	NDX	<i>FROM</i>		
BTC	78.73	21.27	21.27		
NDX	16.31	83.69	16.31		
<i>TO</i>	16.31	21.27			
<i>NET</i>	-4.96	4.96			
<i>TCI</i>					18.79

<i>Panel B: BTC-AI-equity system</i>					
	BTC	NVDA	GOOGL	MSFT	<i>FROM</i>
BTC	74.93	8.63	5.42	11.03	25.07
NVDA	4.20	75.10	10.49	10.21	24.90
GOOGL	6.83	16.59	68.68	7.89	31.32
MSFT	7.59	13.78	7.07	71.56	28.44
<i>TO</i>	18.62	39.01	22.98	29.13	
<i>NET</i>	-6.46	14.11	-8.33	0.68	
<i>TCI</i>					27.43

<i>Panel C: BTC-comparison-equity system</i>					
	BTC	CSCO	BKNG	MDLZ	<i>FROM</i>
BTC	76.45	3.65	18.06	1.84	23.55
CSCO	3.20	78.86	10.23	7.71	21.14
BKNG	6.43	3.89	84.91	4.77	15.09
MDLZ	1.36	3.59	11.85	83.20	16.80
<i>TO</i>	11.00	11.14	40.13	14.32	
<i>NET</i>	-12.55	-10.00	25.04	-2.48	
<i>TCI</i>					19.15

Notes: The table reports average connectedness measures from the TVP-VAR model using forecast horizon $H = 10$. Rows indicate the receivers, columns indicate transmitters. Diagonal elements report own-volatility shares, while off-diagonal elements report cross-variable volatility spillovers. All values are expressed in percentage points.

Within the BTC-AI-equity system, NVIDIA is the dominant net transmitter, with a net

connectedness value of 14.11%. Bitcoin and Alphabet are net receivers, while Microsoft is approximately balanced. These results indicate that volatility transmission is mainly driven by NVIDIA, whereas Bitcoin tends to absorb volatility shocks from AI-exposed equities rather than transmit them. This supports the broader literature documenting volatility spillovers between cryptocurrency and equity markets, although the dominant direction of transmission differs from studies that identify cryptocurrencies as net transmitters, such as Akyildirim et al. (2025).

Figure 2 confirms the same pattern from a pairwise perspective. In the BTC–NDX system, volatility transmission runs mainly from the NASDAQ-100 to Bitcoin. In the BTC–AI-equity system, NVIDIA has the strongest outgoing spillovers, particularly toward Alphabet and Microsoft, while Bitcoin is primarily positioned as a receiver.

Table 5 reports Bitcoin’s average pairwise net directional connectedness with each counterpart. Bitcoin receives net spillovers from NDX (4.96 percentage points), NVIDIA (4.43) and Microsoft (3.44), and transmits a smaller net amount to Alphabet (1.41). The three AI-system components sum to Bitcoin’s net connectedness of -6.46 percentage points.

Table 5: Bitcoin-Specific Net Pairwise Connectedness, $H = 10$

System	Counterpart	Net transmission from BTC
BTC–NDX	NDX	-4.96
BTC–AI equities	GOOGL	1.41
BTC–AI equities	MSFT	-3.44
BTC–AI equities	NVDA	-4.43
BTC–comparison equities	BKNG	-11.62
BTC–comparison equities	CSCO	-0.46
BTC–comparison equities	MDLZ	-0.47

Notes: Means over 501 dates, in percentage points. Positive: net transmission from BTC; negative: net reception.

Figure A2 shows that BTC–AI-equity connectedness is highest in April–June 2024, coinciding with the NVIDIA-led AI rally following its May 2024 earnings surprise and stock-split announcement. Connectedness then declines into early 2025, and the January

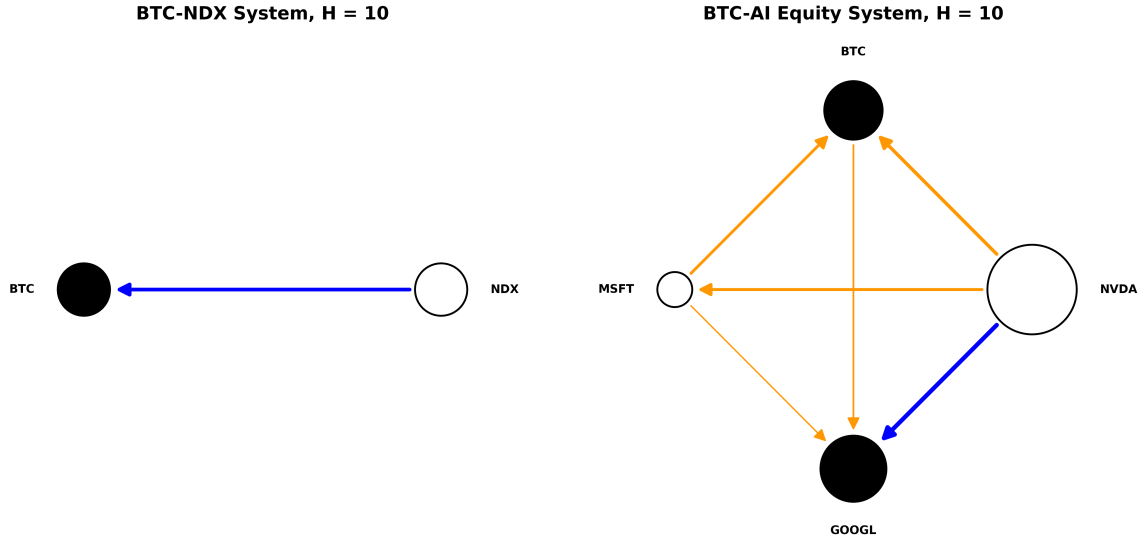


Figure 2: Net Pairwise Directional Connectedness Networks, $H = 10$

Notes: The figure reports average net pairwise directional connectedness from the TVP-VAR model with forecast horizon $H = 10$. Arrows indicate the dominant direction of net volatility spillovers, and edge thickness reflects spillover strength. Node size reflects the absolute strength of net total directional connectedness. A black node color indicates a net recipient and white nodes indicate net transmitters.

2025 DeepSeek-related NVIDIA sell-off does not appear to generate a persistent increase in system-wide connectedness. A broader cross-market episode appears around April 2025, when both systems display higher connectedness and Bitcoin becomes a stronger net receiver, coinciding with the broad risk-off reaction to U.S. tariff announcements. Finally, the late-2025 increase in connectedness is accompanied by a temporary reversal in average net roles, with Bitcoin becoming a net transmitter and NVIDIA weakening as a net transmitter. Because these event links are descriptive, they should be interpreted as contextual rather than causal.

The next subsection examines whether these connectedness measures are systematically related to AI sentiment.

5.3 Main connectedness regressions

This subsection examines whether AI-related sentiment explains variation in volatility connectedness after controlling for broader macro-financial conditions. The dependent variables are the *TCI*, *TO* and *FROM* connectedness measures for both systems, derived using forecast horizon $H = 10$. The main specification uses both the contemporaneous and lagged expectation-adjusted AI sentiment index, EAIS. Because the connectedness measures are estimated time-series objects and may exhibit serial dependence, inference is based on HAC standard errors with a maximum lag length of five.

Table 6 shows that the relation between EAIS and total connectedness differs across the two systems. In the BTC–NDX system, contemporaneous EAIS is negatively associated with total connectedness, but the coefficient is only weakly significant at the 10% level. A one-unit increase in EAIS is associated with a 1.24 percentage-point decline in total connectedness, equivalent to about 6.6% of the system’s average TCI of 18.79%. The lagged EAIS coefficient is small and statistically insignificant. This suggests that there is only limited evidence that AI-related sentiment is associated with total connectedness in the broad BTC–NDX system.

In contrast, the BTC–AI-equity system shows a strongly significant and positive contemporaneous association. A one-unit increase in EAIS is associated with a 2.14 percentage-point increase in total connectedness, equal to about 7.8% of the system’s average TCI of 27.43%. The lagged coefficient is negative and also statistically significant at the 1% level: a one-unit increase in lagged EAIS is associated with 1.98 percentage points lower connectedness. This suggests that the sentiment association with firm-level AI-equity connectedness is strong on the same trading day, with an opposite-signed coefficient on lagged sentiment. The explanatory power is also substantially higher in the AI-equity system, with an adjusted R^2 of 0.324 compared with 0.030 for the benchmark system, indicating that AI sentiment is more informative for firm-level AI-equity connectedness than for broad BTC–technology index connectedness.

The directional regressions in Tables 7 and 8 describe the directional associations. In the BTC–NDX system, a one-unit increase in EAIS is associated with 3.45 percentage points lower Bitcoin transmitted connectedness, which is statistically significant at the 5% level

Table 6: Contemporaneous and lagged EAIS and Total Connectedness

	BTC–NDX system	BTC–AI-equity system
Constant	13.720** (6.570)	53.919*** (4.621)
EAIS	−1.238* (0.708)	2.135*** (0.410)
L. EAIS	0.099 (0.572)	−1.979*** (0.462)
DXY (Log Return)	−3.630 (42.323)	−14.210 (35.781)
VIX	−0.094 (0.073)	0.178** (0.073)
$\ln(GPR)$	0.368 (0.935)	−1.617*** (0.615)
$\ln(EPU)$	0.846 (0.873)	−4.059*** (0.596)
N	487	487
R^2	0.042	0.332
Adjusted R^2	0.030	0.324

Notes: HAC standard errors with maxlags = 5 are reported in parentheses. The dependent variable is the TCI measure. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

and is large relative to its average transmitted connectedness of 16.31%. By comparison, Bitcoin’s received connectedness increases by 0.97 percentage points, but this effect is statistically insignificant. The lagged EAIS coefficients are also insignificant. Thus, for the benchmark system, the evidence mainly supports a contemporaneous reduction in Bitcoin’s role as a volatility transmitter, while evidence for a stronger receiver role is weaker.

The same pattern is stronger and more precisely estimated in the BTC–AI-equity system. A one-unit increase in EAIS is associated with 5.57 percentage points higher Bitcoin received connectedness and 3.15 percentage points lower transmitted connectedness. The corresponding transmitted-connectedness coefficients are 6.08 for NVIDIA, 9.23 for Alphabet and −3.63 for Microsoft; Microsoft’s received-connectedness coefficient is 7.24. Overall, the contemporaneous estimates suggest that elevated AI sentiment is associated with a shift

in volatility dominance away from Bitcoin and toward AI-exposed technology firms, particularly NVIDIA and Alphabet.

Table 7: Contemporaneous and lagged EAIS and Directional Connectedness: BTC–NDX System

	Bitcoin		NASDAQ-100	
	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>
Constant	2.934 (8.505)	24.507*** (7.262)	24.507*** (7.262)	2.934 (8.505)
EAIS	−3.445** (1.407)	0.969 (0.592)	0.969 (0.592)	−3.445** (1.407)
L. EAIS	−0.905 (0.797)	1.104 (0.682)	1.104 (0.682)	−0.905 (0.797)
DXY (Log Return)	−47.071 (60.493)	39.811 (58.594)	39.811 (58.594)	−47.071 (60.493)
VIX	−0.471*** (0.098)	0.282*** (0.109)	0.282*** (0.109)	−0.471*** (0.098)
$\ln(GPR)$	2.077 (1.356)	−1.341 (0.995)	−1.341 (0.995)	2.077 (1.356)
$\ln(EPU)$	2.023 (1.506)	−0.330 (0.831)	−0.330 (0.831)	2.023 (1.506)
<i>N</i>	487	487	487	487
<i>R</i> ²	0.157	0.125	0.125	0.157
Adjusted <i>R</i> ²	0.146	0.115	0.115	0.146

Notes: HAC standard errors with maxlags = 5 are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

The lagged EAIS coefficients in the BTC–AI-equity system qualify this interpretation. Lagged sentiment is negative and statistically significant for Bitcoin’s transmitted connectedness, weakly significant for Bitcoin’s received connectedness, and negative for several firm-level connectedness measures. In particular, lagged EAIS is significantly negative for NVIDIA’s received connectedness and Microsoft’s received connectedness, while the effects for Alphabet are weaker and significant only at the 10% level. The lagged specification therefore supports the view that AI sentiment has a strong same-day association with

spillover direction, with opposite-signed lagged associations conditional on contemporaneous sentiment.

Table 8: Contemporaneous and lagged EAIS and Directional Connectedness: AI-Equity System

	Bitcoin		NVIDIA		Alphabet		Microsoft	
	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>
Constant	31.004*** (10.479)	72.170*** (11.119)	79.730*** (10.520)	34.841*** (8.301)	101.753*** (9.982)	12.976* (7.794)	3.189 (10.530)	95.688*** (7.370)
EAIS	−3.147** (1.258)	5.570*** (0.864)	6.079*** (1.180)	−3.246*** (0.665)	9.232*** (1.092)	−1.028 (0.666)	−3.627*** (0.944)	7.241*** (1.132)
L. EAIS	−5.634*** (1.016)	−1.649* (0.927)	−1.351 (1.111)	−1.613** (0.688)	−1.803* (0.974)	−1.322* (0.783)	0.872 (0.970)	−3.332*** (0.780)
DXY (Log Return)	−26.430 (76.603)	36.019 (80.029)	51.936 (80.982)	−18.354 (54.210)	−103.985 (73.377)	32.409 (52.870)	21.640 (86.922)	−106.912* (58.756)
VIX	−0.309** (0.147)	0.081 (0.117)	0.570*** (0.145)	−0.200* (0.115)	0.337* (0.172)	0.490*** (0.097)	0.112 (0.146)	0.340*** (0.130)
$\ln(GPR)$	1.624 (1.808)	−1.393 (1.484)	−2.694** (1.361)	−2.900* (1.578)	−6.273*** (1.258)	2.501** (1.181)	0.873 (1.658)	−4.677*** (1.129)
$\ln(EPU)$	−2.949** (1.376)	−7.709*** (1.197)	−6.980*** (1.749)	1.512** (0.747)	−9.963*** (1.324)	−0.617 (0.919)	3.658*** (1.301)	−9.420*** (1.160)
N	487	487	487	487	487	487	487	487
R^2	0.293	0.490	0.317	0.253	0.560	0.196	0.154	0.500
Adjusted R^2	0.284	0.484	0.308	0.244	0.555	0.185	0.143	0.494

Notes: HAC standard errors with maxlags = 5 are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

The control variables indicate that macro-financial conditions also matter, but their relevance differs across systems. In the BTC–NDX system, VIX is significant in the directional regressions but not in the total connectedness regression, while $\ln(GPR)$ and $\ln(EPU)$ are statistically insignificant across the reported benchmark-system regressions. In the BTC–AI-equity system, VIX, $\ln(GPR)$ and $\ln(EPU)$ are relevant in several specifications, especially for firm-level transmitted and received connectedness. DXY returns are insignificant in nearly all connectedness regressions, with only weak evidence for Microsoft’s received connectedness in the AI-equity system. The relatively low adjusted R^2

values in the BTC–NDX system indicate that there may still be omitted variables in those regressions.

Overall, the connectedness regressions suggest that AI-related sentiment is most relevant for firm-level AI-equity spillovers. The main result is not a uniform positive association between AI sentiment and connectedness, but differences in the direction and timing of the estimated associations. Contemporaneously, Bitcoin becomes more of a receiver, while NVIDIA and Alphabet become stronger transmitters. Moreover, the opposite-signed lagged coefficients suggest a short-run pattern in the conditional associations; they do not establish a causal correction following a sentiment shock. The evidence for the broad BTC–NDX system is more limited, while the firm-level BTC–AI-equity results provide stronger support for the interpretation that AI-related sentiment is associated with volatility transmission between Bitcoin and AI-exposed equities.

As an additional check on timing, Table A8 reports one-trading-day Granger tests of EAIS on TCI, TO and FROM connectedness, conditional on one own lag. The tests do not reject the absence of predictive content for TCI in either BTC–NDX ($p = 0.846$) or BTC–AI equities ($p = 0.449$). One of the 14 directional and total-connectedness tests rejects at 5%: Bitcoin received connectedness in the AI system ($p = 0.012$). The tests complement the main regressions but do not identify a causal behavioral mechanism. In particular, sentiment and markets may react to the same underlying news, so the same-day coefficients are interpreted as conditional associations.

5.4 Robustness Checks

This subsection evaluates whether the main findings are robust to alternative sentiment measures, a longer connectedness forecast horizon, dynamic sentiment specifications in the connectedness regressions, and including sentiment as an endogenous variable.

5.4.1 Alternative sentiment measure in the ARMA-EGARCH

The ARMA-EGARCH models are re-estimated using raw AI sentiment, while keeping the benchmark lag orders unchanged. As seen in Table A9, the raw sentiment coefficients are statistically insignificant in both the mean and variance equations for NASDAQ-100.

Moreover, the estimation for BTC with AIS is unstable, because of the strong time-trend in AIS, which limits the interpretation of this robustness check.

5.4.2 Alternative forecast horizon in the TVP-VAR

The connectedness analysis is repeated using a longer forecast horizon of $H = 100$. As shown in Table A10, total connectedness increases in both systems with a longer horizon. Directionally, the results are similar to $H = 10$, though Microsoft becomes a more pronounced net transmitter at the longer horizon. Overall, the $H = 100$ results confirm that the BTC–AI-equity system displays stronger volatility interdependence than the broad BTC–NDX benchmark system.

5.4.3 Alternative sentiment specification in the connectedness regressions

As an additional robustness check, the connectedness regressions are estimated using the AIS instead of the EAIS. This specification assesses whether the sentiment-connectedness relationship is driven by the unfiltered AI sentiment index or whether the orthogonalized EAIS provides additional information. Statistical inference is based on HAC standard errors with five lags.

The results in Table A11 show that AIS is not significantly associated with total connectedness in the BTC–NDX system. In the BTC–AI-equity system, the contemporaneous AIS coefficient is also statistically insignificant, while the lagged AIS coefficient is positive and significant. This differs from the main EAIS specification, where contemporaneous sentiment is positively associated with total connectedness in the AI-equity system and the lagged coefficient is negative. Hence, the raw sentiment measure does not reproduce the same short-run reversal pattern found for EAIS. Instead, it suggests that the unadjusted sentiment index captures a more delayed and less precisely identified form of AI-related market attention.

The directional regressions in Tables A12 and A13 provide only partial support for the main directional interpretation. In the BTC–NDX system, contemporaneous AIS is insignificant across all directional connectedness measures, while lagged AIS is positive and significant for Bitcoin’s received connectedness and the NASDAQ-100’s transmitted

connectedness. In the BTC–AI–equity system, contemporaneous AIS weakly reduces Bitcoin’s transmitted connectedness, but most contemporaneous coefficients are statistically insignificant. The lagged AIS coefficients are positive and significant for Bitcoin’s received connectedness and Alphabet’s transmitted connectedness, suggesting that unadjusted sentiment is associated with a limited set of delayed directional spillover effects in the AI–equity network.

Overall, the AIS results provide weaker and less consistent evidence than the EAIS results. The evidence is less systematic than in the EAIS specification, indicating that the main findings depend on how the sentiment measure is constructed.

5.4.4 Including EAIS endogenously in the TVP-VAR

As a final robustness check, the TVP-VAR connectedness analysis is estimated after including expectation-adjusted AI sentiment as an endogenous variable in the system. Treating EAIS endogenously allows sentiment to have a dynamic impact by transmitting and receiving shocks within the system.

Table A14 reports the average connectedness tables for the BTC–NDX–EAIS and BTC–AI–equity–EAIS systems using $H = 10$. In the BTC–NDX–EAIS system, EAIS is a net transmitter, Bitcoin remains the main net receiver, while the NASDAQ-100 is close to balanced.

In the BTC–AI–equity–EAIS system, the overall structure is broadly consistent with the baseline firm-level connectedness results. NVIDIA remains a net transmitter, while Bitcoin and Alphabet remain net receivers. EAIS is a net transmitter, with a NET of 2.54 percentage points, and most of its forecast-error variance remains self-explained. This indicates that EAIS does not dominate the volatility system when modeled endogenously. Instead, it acts as a relatively independent sentiment node that transmits shocks to financial variables in specific periods.

The corresponding network and total-connectedness Figures in A5, A6 and A7 support this interpretation. The inclusion of EAIS adds additional sentiment-related transmission, but it does not overturn the main finding that firm-level AI–equity connectedness is stronger and more informative than broad BTC–NDX connectedness. Moreover, the total connect-

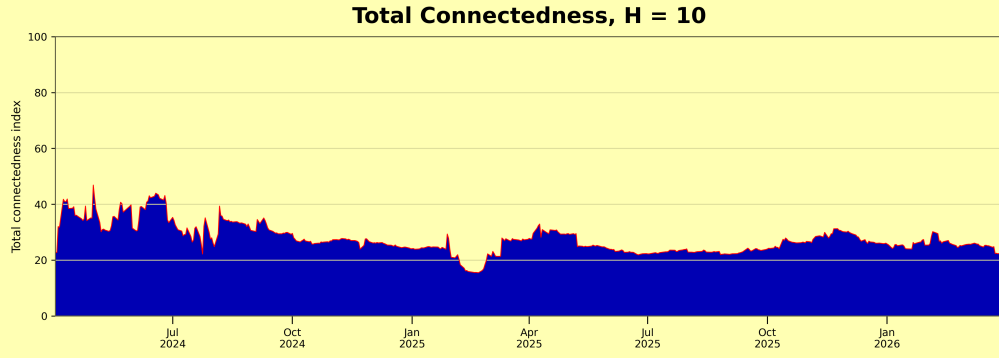
edness plots show that the role of EAIS is time-varying and concentrated during periods of heightened attention.

5.4.5 Same-size comparison system: Cisco, Booking and Mondelez

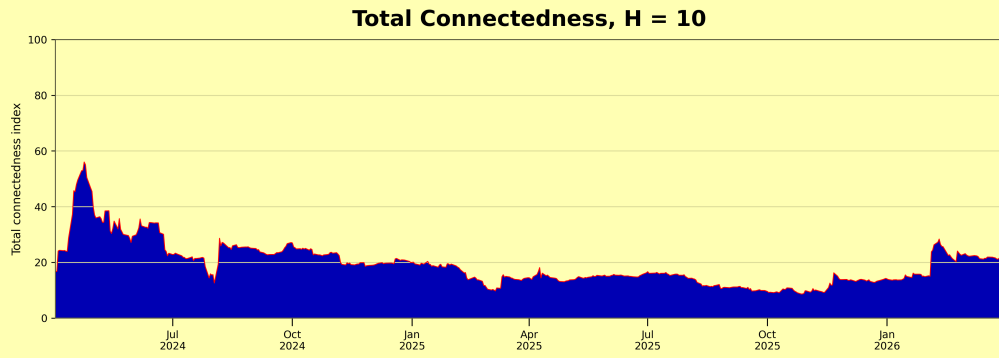
To assess whether the higher connectedness in the BTC–AI-equity system reflects system size, we estimate an equally sized system consisting of BTC, CSCO, BKNG and MDLZ. The comparison firms span different degrees of sectoral proximity to AI. A case-insensitive search of the 92,407 in-sample texts identifies 34 Cisco documents, seven Booking documents and four Mondelez documents, compared with 2,167 for NVIDIA, 7,464 for Google and Alphabet, and 3,775 for Microsoft. The search uses company names and tickers and it is important to note that this is a measure of narrative salience, not a comprehensive measure of AI business exposure. The firms are not matched in market capitalization or sector, and Cisco has AI-infrastructure exposure, so the comparison does not isolate AI exposure from every other firm characteristic.

We estimate the three additional EGARCH(1,1) volatility models and the TVP-VAR using the same procedure as in Sections 4.1–4.2, retaining the existing Bitcoin volatility series. All three EGARCH fits converge over 501 trading-day returns. BIC selects TVP-VAR(1); the forecast horizon is $H = 10$ and both forgetting factors are 0.99.

Panel C of Table 4 reports average comparison-system TCI of 19.15%, close to 18.79% for BTC–NDX and below 27.43% for the equally sized AI-equity basket. Thus, increasing the system to four variables does not by itself reproduce the higher connectedness observed for the AI equities. The result addresses the system-size concern, while differences in firm and sector composition remain relevant. Bitcoin is a net receiver in the comparison system (−12.55 percentage points), predominantly from Booking (Table 5). Figure 3 compares the two four-variable TCI paths.



(a) BTC-AI-equity system



(b) BTC-comparison-equity system

Figure 3: Total Connectedness in Equally Sized Systems, $H = 10$

Notes: Both systems contain BTC and three equities; the comparison basket comprises CSCO, BKNG and MDLZ.

Although average connectedness is lower for the comparison basket, the sentiment association is not absent from it. The comparison therefore supports the conclusion that the higher AI-basket TCI is not merely a consequence of system size, but it does not support interpreting the sentiment-connectedness association as exclusive to the selected AI firms.

Table 9: EAIS and Total Connectedness in Equally Sized Systems

	AI equities	Comparison equities
Constant	53.919*** (4.621)	47.223*** (7.047)
EAIS	2.135*** (0.410)	4.636*** (0.949)
L. EAIS	-1.979*** (0.462)	-2.296*** (0.656)
DXY (Log Return)	-14.210 (35.781)	22.493 (46.970)
VIX	0.178** (0.073)	0.079 (0.073)
$\ln(GPR)$	-1.617*** (0.615)	1.441 (1.210)
$\ln(EPU)$	-4.059*** (0.596)	-6.889*** (0.859)
N	487	487
R^2	0.332	0.467
Adjusted R^2	0.324	0.460

Notes: HAC standard errors in parentheses. ***, ** and * denote 1%, 5% and 10% significance.

6 Discussion & Conclusion

This study examines a broader question in behavioral finance and portfolio management: whether investment narratives can alter cross-market risk transmission and thereby affect the diversification properties of multi-asset portfolios. Using the emergence of AI as a prominent investment narrative, the analysis combines text-based sentiment measures with asset-level volatility models and dynamic connectedness techniques to examine the relationship between narrative-driven sentiment and risk transmission across Bitcoin, broad technology equities, and selected AI-exposed firms. The results provide evidence that the

relevance of investor sentiment is not confined to its association with individual asset returns: narrative-linked sentiment is also associated with the structure and direction of volatility transmission across markets.

The ARMA-EGARCH results provide evidence of a sentiment-return relationship for Bitcoin. Higher expectation-adjusted AI sentiment is associated with higher next-day Bitcoin returns, while the corresponding effect for the NASDAQ-100 is positive but statistically insignificant. In contrast, AI sentiment does not have a statistically significant direct effect on conditional volatility for either Bitcoin or the NASDAQ-100. This differs from the volatility effects documented by Guler (2021), and suggests that in this setting the relevance of AI sentiment is more visible in Bitcoin returns and cross-market transmission than in standalone conditional variance.

The TVP-VAR results reveal important differences in the way risk is transmitted across broad and thematic market exposures. Volatility spillovers are stronger in the BTC–AI–equity system than in the broader BTC–NASDAQ-100 system, indicating that a broad technology index may understate the degree of connectedness between Bitcoin and firms that are more directly exposed to a dominant investment narrative. At the firm level, NVIDIA emerges as the dominant net transmitter, while Bitcoin and Alphabet are net receivers on average and Microsoft is approximately balanced. These findings suggest that conventional asset-class and broad-index classifications may conceal important firm-level sources of portfolio risk. For institutional investors, this suggests that aggregate asset-class monitoring may provide an incomplete picture when portfolios contain securities exposed to a common investment theme.

The same-size comparison in Section 5.4.5 supports this connectedness ranking: its average TCI is 19.15%, below 27.43% for the AI-equity basket. However, its sentiment coefficient is also positive and significant. The comparison addresses the mechanical system-size concern, while showing that the sentiment association is not exclusive to the selected AI firms.

The central finding of the analysis is that narrative-driven sentiment is not uniformly associated with higher cross-market connectedness; instead, its associations differ across the structure and direction of risk transmission. Expectation-adjusted AI sentiment is nega-

tively associated with total connectedness in the BTC–NASDAQ-100 system but positively associated with connectedness in the BTC–AI-equity system. This contrast suggests that the association with an investment narrative can differ across the economic and thematic exposure of the assets involved. In particular, elevated AI sentiment is associated with a stronger risk linkage between Bitcoin and firms that are more directly exposed to the AI investment theme, without a corresponding positive association for Bitcoin and the broader technology market.

The directional results provide further evidence of this reallocation of risk transmission: higher AI sentiment is associated with a smaller transmitter role and a larger receiver role for Bitcoin, alongside stronger transmission from NVIDIA and Alphabet. These findings suggest that narrative-driven sentiment is associated with differences in the architecture of cross-market risk and potentially concentrated sources of portfolio exposure that are not visible when investors rely only on broad asset-class or sector-level measures of connectedness. From an institutional perspective, therefore, the results highlight the importance of monitoring thematic and firm-level exposures alongside traditional measures of diversification.

The Granger results in Table A8 provide limited evidence that EAIS precedes connectedness: neither TCI test rejects, and the directional rejection is isolated among 14 tests. Thus, the directional patterns remain relevant to describing cross-market risk, but do not by themselves demonstrate that sentiment causes its transmission. The portfolio implications are interpretations of the observed associations.

The robustness checks qualify the earlier interpretation, particularly because the raw-sentiment specifications do not consistently reproduce the main findings. When raw AI sentiment is used in the ARMA-EGARCH models, the NASDAQ-100 mean and variance coefficients are insignificant and the BTC specification is unstable, showing sensitivity to filtering sentiment using prediction-market and macro-financial information. Extending the connectedness forecast horizon from 10 to 100 days increases the overall level of connectedness, but leaves the main directional interpretation largely unchanged. The alternative AIS regressions provide weaker and less stable evidence than the EAIS specification, while the endogenous-EAIS robustness check shows that AI sentiment is not a stable dominant

transmitter. Overall, the results suggest that AI sentiment is best interpreted as a temporary and state-dependent narrative channel, rather than as a persistent source of volatility transmission.

The divergence between AIS and EAIS deserves separate consideration. Table A3 shows that raw AI sentiment contains variation associated with macro-financial, geopolitical and policy-uncertainty conditions. Removing that variation can therefore change the estimated relationship with connectedness. This helps explain why the two measures need not yield the same results, but does not validate EAIS as a purified speculative component. The main findings should be understood as conditional on the specified filtering procedure rather than as a general property of every measure of AI-related sentiment.

The findings have several implications for institutional investors and risk managers. First, diversification should be viewed as a dynamic rather than static property. Historical correlations and average connectedness measures may not fully capture the risk of portfolios during periods when a dominant investment narrative changes investor behavior and positioning. Second, investors should consider thematic and behavioral exposures alongside conventional asset-class classifications. The results show that Bitcoin and broad technology equities do not necessarily become more connected during periods of elevated AI sentiment, yet Bitcoin becomes more closely linked to individual AI-exposed firms, a divergence that index-level risk monitoring would have missed entirely. Third, sentiment and narrative indicators may provide useful supplementary information for dynamic risk monitoring and stress testing. A concrete practice suggested by these results is to construct and monitor narrow, theme-based baskets (e.g., AI-exposed equities, ESG-flagged assets, or other thematically linked groups) alongside standard asset-class buckets, and to use elevated thematic sentiment as a trigger for targeted stress tests of cross-asset exposure within those baskets, rather than relying solely on periodic, aggregate correlation reviews. More broadly, the findings suggest that investors should not assume that assets classified as belonging to different asset classes will continue to provide independent sources of risk when market participants become collectively focused on the same investment narrative, a dynamic we expect to recur around future investment narratives beyond AI.

It is worth being explicit about the scope of these claims. The analysis is based on

one narrative episode and one pair of asset classes, and the results should not be read as evidence that AI sentiment specifically, or cryptocurrency markets generally, are uniquely exposed to narrative-driven connectedness. Rather, the AI–Bitcoin setting serves as a case study of a broader mechanism: when investor attention concentrates on a specific theme, the assets most closely tied to that theme can become more tightly linked in their volatility dynamics even while broader, index-level measures show little change. We would expect analogous patterns around other salient investment narratives and see testing this mechanism in other thematic settings as a natural direction for future research.

Several limitations should be considered when interpreting these findings. First, the sample covers a relatively short post-ChatGPT period. This limits the number of observations, but it is also necessary for studying a period in which generative AI was a market-wide narrative and prediction-market data were available. Second, the expectation-adjusted sentiment index is only a proxy for speculative AI sentiment. Although the orthogonalization procedure filters out observable expectation and macro-financial components, the residual may still contain omitted variables and should not be interpreted as a pure measure of irrational beliefs. Third, the analysis is based on daily data, which makes it possible to combine textual sentiment, prediction-market variables, macro-financial controls, and asset prices, but may miss intraday reactions to AI-related news. Finally, the relatively low adjusted R^2 values indicate that the models explain only part of the variation in connectedness. Combined with the observational nature of the analysis, this means that the results should be interpreted as conditional associations rather than causal effects.

A further limitation concerns the roughly two-year sample. It reflects both the selected post-ChatGPT period and the availability of the Kalshi series from April 2024 onward. The effective sample contains 501 return observations and 487 observations in the main lagged connectedness regressions. The TVP-VAR is estimated using Kalman filtering with forgetting factors, following Koop and Korobilis (2014), so it does not require a fixed rolling-window length. This does not remove the limitations of a short sample or guarantee statistical power. A longer sample would help assess the stability of the estimates across episodes and increase the information available for inference.

Future research could extend this study in three main directions. First, a quantile-

based connectedness framework could test whether AI sentiment becomes more important during extreme market conditions. Second, higher-frequency data could be used to examine whether AI sentiment affects volatility transmission intraday, especially around major AI announcements. Finally, a broader asset network could help determine whether the identified sentiment channel is specific to Bitcoin and US AI-exposed technology firms, or whether it reflects a wider cross-market AI narrative.

In conclusion, expectation-adjusted AI sentiment is associated with cross-market volatility connectedness in a directional and short-lived manner. The evidence does not indicate a uniform increase in Bitcoin–NASDAQ-100 connectedness. Instead, the relationship is concentrated in the firm-level system containing Bitcoin and AI-exposed equities, particularly through changes in the transmitted connectedness of NVIDIA and Alphabet and the received connectedness of Bitcoin. These findings suggest that broad-index measures may provide an incomplete view of thematic cross-asset exposure. For portfolio and risk managers, theme-based connectedness and sentiment measures may therefore offer useful supplementary inputs for exposure monitoring and stress testing, although the results should be interpreted as conditional associations rather than causal or directly tradable signals.

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A Appendix

A.1 Textual sentiment scoring and daily aggregation

For each text observation k , the continuous sentiment score is defined as

$$s_k = P_k(\text{positive}) - P_k(\text{negative}), \quad (\text{A.1})$$

where higher values indicate more positive AI-related sentiment. The scores are aggregated separately by source l and assigned equity trading day d :

$$S_d^{(l)} = \frac{1}{N_d^{(l)}} \sum_{k=1}^{N_d^{(l)}} s_{k,d}^{(l)}, \quad (\text{A.2})$$

where $N_d^{(l)}$ counts all source- l texts assigned to trading day d , including those from preceding non-trading days.

A.2 Dynamic Factor Model

Let y_t denote the vector of observed source-specific daily sentiment scores. The measurement equation of the one-factor Dynamic Factor Model is

$$y_t = \Lambda f_t + e_t, \quad e_t \sim \mathcal{N}(0, R), \quad (\text{A.3})$$

where f_t is the latent common AI-sentiment factor, Λ contains the factor loadings, and e_t contains source-specific measurement errors. The latent factor follows

$$f_t = \sum_{i=1}^p A_i f_{t-i} + u_t, \quad u_t \sim \mathcal{N}(0, Q). \quad (\text{A.4})$$

A.3 TVP-VAR specification

Let y_t denote an $m \times 1$ vector of endogenous conditional-volatility series, and let z_{t-1} contain their lags. The observation equation is

$$y_t = B_t z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma_t), \quad (\text{A.5})$$

where B_t is the matrix of time-varying coefficients. Its dynamics follow

$$\text{vec}(B_t) = \text{vec}(B_{t-1}) + v_t, \quad v_t \sim \mathcal{N}(0, \Omega_t). \quad (\text{A.6})$$

A.4 Connectedness measures

The time-varying vector moving-average representation is

$$y_t = \sum_{h=0}^{\infty} \Psi_{h,t} \varepsilon_{t-h}. \quad (\text{A.7})$$

The generalized forecast error variance decomposition is

$$\phi_{ij,t}(H) = \frac{\Sigma_{jj,t}^{-1} \sum_{h=0}^{H-1} (e_i' \Psi_{h,t} \Sigma_t e_j)^2}{\sum_{h=0}^{H-1} (e_i' \Psi_{h,t} \Sigma_t \Psi_{h,t}' e_i)}, \quad (\text{A.8})$$

where e_i is the i th unit vector. Its normalized form is

$$\tilde{\phi}_{ij,t}(H) = \frac{\phi_{ij,t}(H)}{\sum_{j=1}^m \phi_{ij,t}(H)}. \quad (\text{A.9})$$

The total connectedness index is

$$TCI_t = 100 \times \frac{1}{m} \sum_{\substack{i,j=1 \\ i \neq j}}^m \tilde{\phi}_{ij,t}(H). \quad (\text{A.10})$$

The directional connection transmitted by the variable i is

$$TO_{i,t} = 100 \times \sum_{\substack{j=1 \\ j \neq i}}^m \tilde{\phi}_{ji,t}(H), \quad (\text{A.11})$$

while the directional connectedness received by variable i is

$$FROM_{i,t} = 100 \times \sum_{\substack{j=1 \\ j \neq i}}^m \tilde{\phi}_{ij,t}(H). \quad (\text{A.12})$$

Net connectedness is

$$NET_{i,t} = TO_{i,t} - FROM_{i,t}, \quad (\text{A.13})$$

where positive values identify net transmitters and negative values identify net receivers.

Finally, net pairwise directional connectedness is

$$NPDC_{ij,t} = 100 \times \left[\tilde{\phi}_{ji,t}(H) - \tilde{\phi}_{ij,t}(H) \right]. \quad (\text{A.14})$$

A.5 Supplementary tables and figures

Table A1: Text Filtering Keyword Lexicon

Category	Exact Search Terms
Broad & Technical Terms	artificial intelligence, machine learning, deep learning, ai model, ai models, ai-generated, agi
Generative & Language Modeling	large language model, large language models, llm, llms, generative ai, genai
Firms & Organizations	openai, anthropic
Specific Models & Chatbots	chatgpt, claude, gemini, bard, copilot, ai chatbot, ai chatbots

Table A2: Manual Validation of RoBERTa Sentiment Classification

Manual label	Model label			Total
	Positive	Neutral	Negative	
Positive	15	16	0	31
Neutral	0	48	1	49
Negative	2	9	9	20
Total	17	73	10	100

Notes: The table compares manual labels with the discrete RoBERTa model labels for 100 manually reviewed observations. The sample consists of 50 GDELT headlines and 50 Reddit posts.

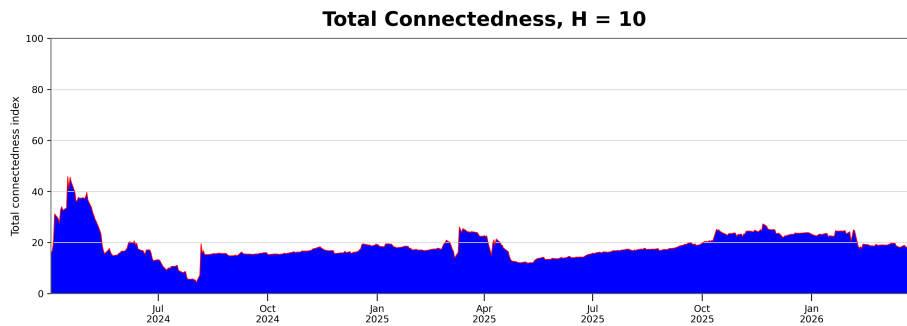


Figure A1: Total connectedness (TCI): BTC-NDX system

Table A3: Orthogonalization Regression Results for the AIS

Dependent Variable: AIS	
Constant	6.462*** (0.471)
Kalshi (Diff)	0.000 (0.016)
Metaculus (Diff)	−0.000 (0.000)
S&P 500 (Log Return)	1.816 (2.978)
VIX (Close)	0.019*** (0.007)
$\ln(EPU)$	−0.856*** (0.071)
DXY (Log Return)	−2.729 (7.479)
$\ln(GPR)$	−0.451*** (0.107)
N	488
R^2	0.494
Adjusted R^2	0.486

Notes: Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

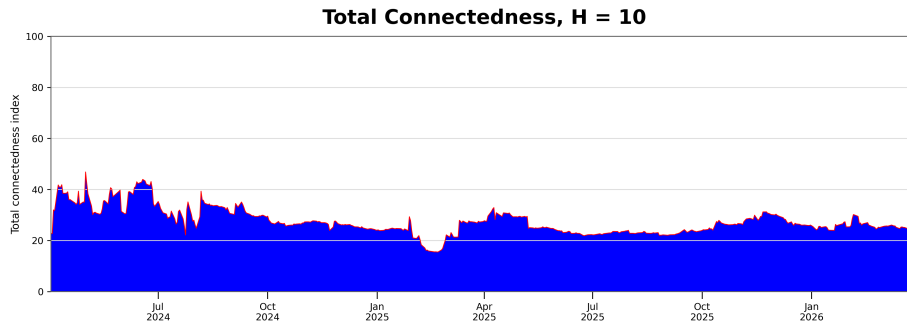


Figure A2: Total connectedness (TCI): BTC–AI-equity system

Table A4: Dynamic Factor Model Selection and Factor Loadings

Lag order	BIC	Log-likelihood	BIC-selected
1	2307.536	-1135.112	Yes
2	2311.981	-1134.225	No
3	2313.616	-1131.934	No
4	2315.455	-1129.744	No
5	2320.376	-1129.095	No
Source	Factor loading		
GDELT news sentiment	0.443		
Reddit sentiment	0.760		

Notes: The table reports lag selection using BIC. All models converged.

Table A5: EGARCH(1,1) Post-Estimation Diagnostics

Statistic	BTC	NDX	NVDA	GOOGL	MSFT
Log-likelihood	-1230.67	-772.50	-1228.63	-990.53	-851.12
AIC	4.937	3.108	4.929	3.978	3.422
BIC	4.987	3.158	4.979	4.029	3.472
HQIC	4.957	3.128	4.948	3.998	3.441
$Q_z(10)$	8.59	6.92	15.78	4.47	2.11
	(0.571)	(0.733)	(0.106)	(0.924)	(0.995)
$Q_z(20)$	31.19	13.09	23.61	17.33	13.97
	(0.053)	(0.873)	(0.260)	(0.631)	(0.832)
$Q_z^2(10)$	18.09	12.16	3.69	4.34	4.57
	(0.053)	(0.274)	(0.960)	(0.931)	(0.918)
$Q_z^2(20)$	21.55	15.39	8.35	7.04	12.38
	(0.366)	(0.754)	(0.989)	(0.997)	(0.902)

Notes: $N = 501$ for all models. $Q_z(k)$ and $Q_z^2(k)$ denote Ljung-Box test statistics for standardized and squared standardized residuals at lag k , respectively. P-values are reported in parentheses.

Table A6: Descriptive Statistics and Diagnostics of Exogenous Control Variables

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	Jarque-Bera	ADF
Kalshi (Diff)	0.025	2.133	-16.000	13.000	-0.288	14.167	2485.44***	-16.90***
Metaculus (Diff)	0.726	59.718	-607.796	289.361	-2.305	31.176	16230.28***	-22.03***
S&P 500 (Log Return)	0.001	0.010	-0.062	0.091	0.436	18.751	4949.87***	-23.38***
VIX (Close)	18.156	4.942	11.860	52.330	2.442	12.957	2451.73***	-4.91***
$\ln(EPU)$	5.378	0.765	1.685	6.792	-0.779	3.372	51.64***	-2.54
DXY (Log Return)	-0.000	0.004	-0.020	0.016	-0.414	4.731	72.70***	-23.04***
$\ln(GPR)$	5.011	0.371	3.596	6.292	0.225	4.138	29.34***	-7.37***

Notes: $N = 488$ for all series. Diff indicates that the series is first-differenced. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A7: Granger Causality Tests for Expectation-Adjusted AI Sentiment

Null Hypothesis	Lag	F-stat	p -val
EAIS does not Granger-cause BTC returns	1	8.123	0.005
EAIS does not Granger-cause NDX returns	1	1.068	0.302

Notes: The table reports one-day bivariate Granger causality tests. EAIS denotes the expectation-adjusted AI Sentiment Index. The number of observations is $N = 487$.

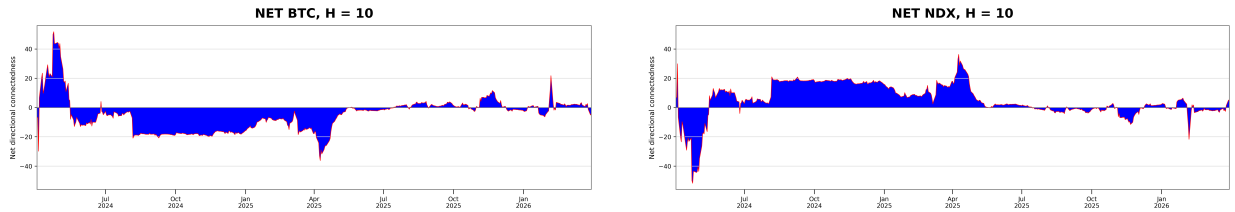


Figure A3: Net total directional connectedness (NET): BTC–NDX system

Table A8: One-Day Granger Tests of EAIS on Connectedness

System	Measure	F	p
BTC–NDX	TCI	0.038	0.846
BTC–NDX	BTC TO	2.053	0.153
BTC–NDX	NDX TO	3.419	0.065
BTC–NDX	BTC FROM	3.419	0.065
BTC–NDX	NDX FROM	2.053	0.153
BTC–AI equities	TCI	0.574	0.449
BTC–AI equities	BTC TO	0.716	0.398
BTC–AI equities	NVDA TO	2.809	0.094
BTC–AI equities	GOOGL TO	0.983	0.322
BTC–AI equities	MSFT TO	0.002	0.969
BTC–AI equities	BTC FROM	6.425	0.012
BTC–AI equities	NVDA FROM	0.058	0.810
BTC–AI equities	GOOGL FROM	0.614	0.434
BTC–AI equities	MSFT FROM	0.000	0.983

Notes: Null: no predictive contribution from EAIS, conditional on one own lag and a constant. $N = 487$; $F(1, 484)$. Unadjusted p -values.

Table A9: Robustness ARMA-EGARCH Estimation Results AIS

	Bitcoin	NASDAQ-100
μ	0.009 (0.022)	0.110** (0.043)
θ_1	-0.078 (0.112)	- -
λ	0.780 (0.789)	0.048 (0.040)
ω	9.128 (14.756)	0.004 (0.014)
α	-0.030 (0.058)	-0.220*** (0.033)
β	0.895 (0.726)	0.947*** (0.023)
γ	1.486 (1.293)	0.060 (0.038)
η	-13.326 (11.002)	0.001 (0.011)
ν	3.332 (6.290)	5.322*** (1.378)
N	500	500
Log-likelihood	-8373.868	-769.524
BIC	33.607	3.178
$Q_z(10)$	0.201	6.344
$Q_z^2(10)$	0.002	9.005
$Q_z(20)$	2.127	12.483
$Q_z^2(20)$	0.008	12.341

Notes: See notes in Table 3

Table A10: Average TVP-VAR Connectedness Table, $H = 100$

<i>Panel A: BTC–NDX system</i>					
	BTC	NDX	<i>FROM</i>		
BTC	75.21	24.79	24.79		
NDX	18.28	81.72	18.28		
<i>TO</i>	18.28	24.79			
<i>NET</i>	−6.51	6.51			
<i>TCI</i>					21.54

<i>Panel B: BTC–AI-equity system</i>					
	BTC	NVDA	GOOGL	MSFT	<i>FROM</i>
BTC	62.78	11.85	7.87	17.50	37.22
NVDA	8.00	61.05	11.01	19.94	38.95
GOOGL	9.54	21.14	55.11	14.21	44.89
MSFT	10.35	16.97	9.07	63.62	36.38
<i>TO</i>	27.88	49.96	27.95	51.65	
<i>NET</i>	−9.34	11.01	−16.94	15.26	
<i>TCI</i>					39.36

Notes: See notes in Table 4, but here $H = 100$.

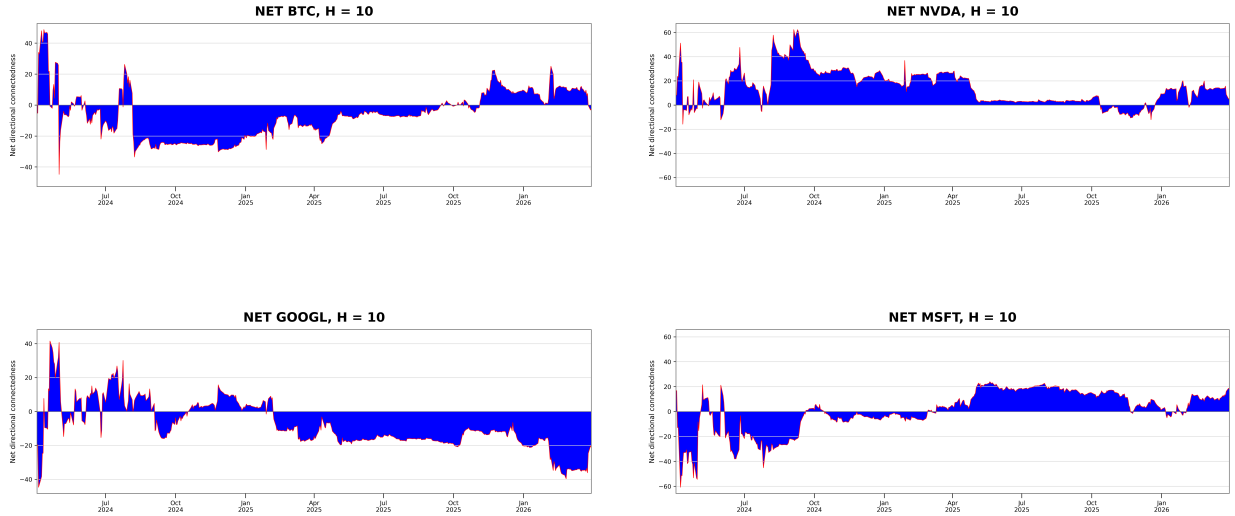


Figure A4: Net total directional connectedness (NET): BTC–AI-equity system

Table A11: Contemporaneous and Lagged AIS and Total Connectedness

	BTC–NDX system	BTC–AI-equity system
Constant	13.142 (9.103)	43.502*** (5.571)
AIS_t	−2.757 (1.887)	−1.990 (1.514)
AIS_{t-1}	2.697 (2.004)	3.180** (1.577)
DXY (Log Return)	29.616 (50.094)	29.352 (39.247)
VIX	−0.085 (0.078)	0.129* (0.072)
$\ln(GPR)$	1.231 (1.470)	−0.796 (0.743)
$\ln(EPU)$	0.188 (0.760)	−2.688*** (0.642)
N	500	500
R^2	0.013	0.319
Adjusted R^2	0.001	0.310

Notes: See Table 6.

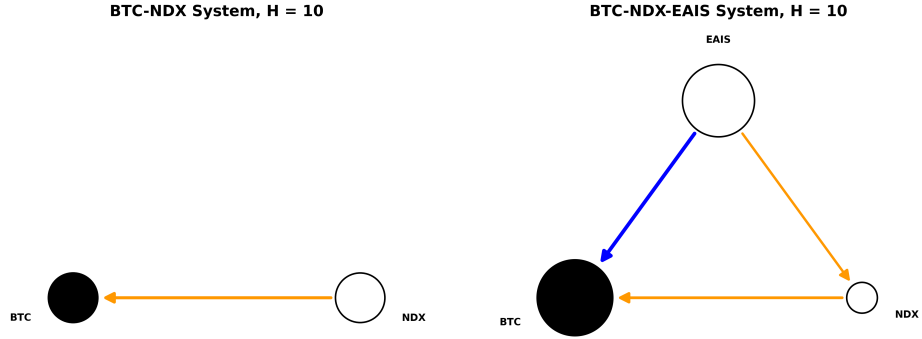


Figure A5: Net pairwise directional connectedness networks for the BTC–NDX benchmark system and the BTC–NDX–EAIS system, $H = 10$.

Table A12: Contemporaneous and Lagged AIS and Directional Connectedness: BTC–NDX System

	Bitcoin		NASDAQ-100	
	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>
Constant	13.048 (14.699)	13.235* (7.497)	13.235* (7.497)	13.048 (14.699)
AIS_t	−3.909 (3.530)	−1.605 (1.871)	−1.605 (1.871)	−3.909 (3.530)
AIS_{t-1}	1.823 (3.921)	3.570** (1.781)	3.570** (1.781)	1.823 (3.921)
DXY (Log Return)	7.122 (78.461)	52.109 (56.474)	52.109 (56.474)	7.122 (78.461)
VIX	−0.441*** (0.119)	0.270*** (0.104)	0.270*** (0.104)	−0.441*** (0.119)
$\ln(GPR)$	2.779 (2.544)	−0.317 (1.017)	−0.317 (1.017)	2.779 (2.544)
$\ln(EPU)$	−0.505 (1.217)	0.882 (0.803)	0.882 (0.803)	−0.505 (1.217)
N	500	500	500	500
R^2	0.072	0.135	0.135	0.072
Adjusted R^2	0.061	0.124	0.124	0.061

Notes: See Table 7.

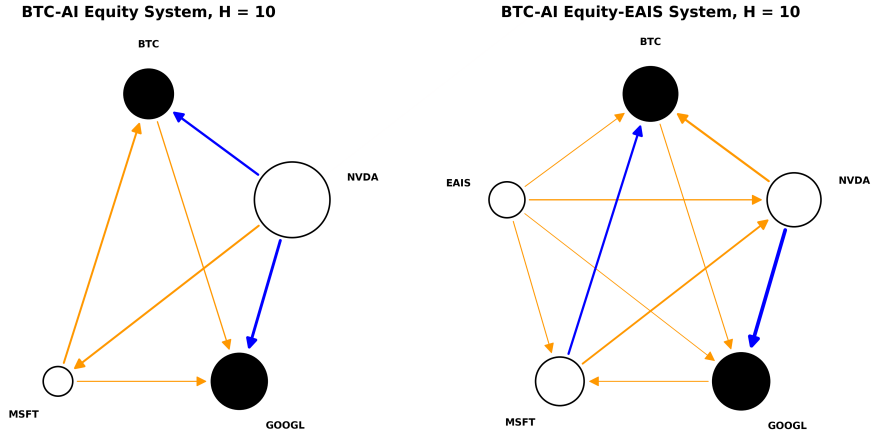


Figure A6: Net pairwise directional connectedness networks for the BTC–AI-equity system and the BTC–AI-equity–EAIS system, $H = 10$.

Table A13: Contemporaneous and Lagged AIS and Directional Connectedness: AI-Equity System

	Bitcoin		NVIDIA		Alphabet		Microsoft	
	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>	<i>TO</i>	<i>FROM</i>
Constant	49.241*** (18.965)	48.470*** (11.104)	49.395*** (10.087)	54.565*** (10.112)	51.962*** (10.374)	18.443* (9.521)	23.411** (10.193)	52.531*** (9.885)
AIS_t	-7.682* (4.083)	-3.436 (2.310)	2.445 (3.519)	-3.329 (2.693)	0.078 (2.941)	-2.587 (2.352)	-2.800 (3.351)	1.392 (3.074)
AIS_{t-1}	3.615 (4.756)	6.875*** (2.368)	2.088 (3.501)	-0.299 (2.629)	7.180** (2.938)	1.583 (2.384)	-0.165 (3.308)	4.559 (3.270)
DXY (Log Return)	97.964 (125.455)	56.100 (77.456)	77.658 (77.322)	-2.808 (56.693)	-55.164 (72.882)	69.709 (61.548)	-3.050 (85.839)	-5.593 (75.918)
VIX	-0.330* (0.170)	0.008 (0.110)	0.467*** (0.141)	-0.156 (0.113)	0.203 (0.163)	0.474*** (0.102)	0.177 (0.144)	0.191 (0.128)
$\ln(GPR)$	1.719 (3.009)	-0.465 (1.511)	-1.189 (1.472)	-3.800** (1.753)	-3.282** (1.411)	2.365* (1.368)	-0.431 (1.592)	-1.285 (1.564)
$\ln(EPU)$	-6.212*** (1.817)	-3.956*** (1.181)	-2.395* (1.433)	-1.457* (0.879)	-3.014*** (1.123)	-1.414 (1.012)	0.867 (1.247)	-3.927*** (1.097)
N	500	500	500	500	500	500	500	500
R^2	0.115	0.439	0.288	0.179	0.525	0.136	0.160	0.483
Adjusted R^2	0.104	0.432	0.279	0.169	0.519	0.125	0.150	0.476

Notes: See Table 8.

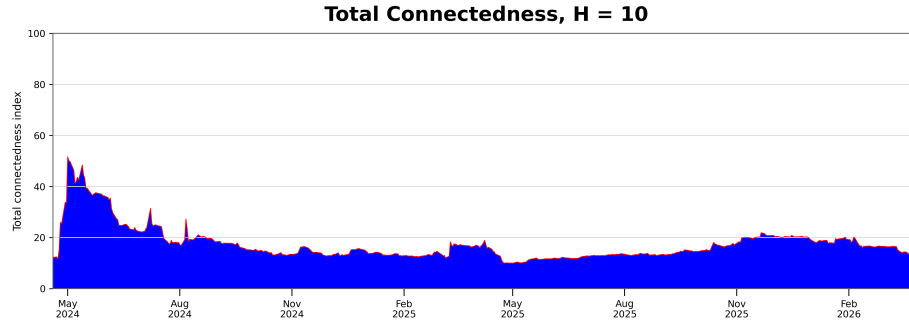


Figure A7: Total connectedness in the BTC–NDX–EAIS system, $H = 10$.

Table A14: Average TVP-VAR Connectedness with Endogenous EAIS, $H = 10$

Panel A: BTC–NDX–EAIS system					
	BTC	NDX	EAIS	FROM	
BTC	72.07	19.83	8.10	27.93	
NDX	15.41	79.30	5.29	20.70	
EAIS	1.73	1.65	96.62	3.38	
TO	17.14	21.48	13.39	52.01	
NET	-10.79	0.78	10.01		
cTCI/TCI				26.01 / 17.34	

Panel B: BTC–AI-equity–EAIS system						
	BTC	NVDA	GOOGL	MSFT	EAIS	FROM
BTC	71.82	9.90	4.26	10.36	3.66	28.18
NVDA	5.88	70.12	6.51	11.28	6.21	29.88
GOOGL	4.94	14.86	68.94	9.01	2.26	31.06
MSFT	6.17	8.13	9.46	73.32	2.91	26.68
EAIS	3.29	4.77	2.19	2.26	87.50	12.50
TO	20.28	37.66	22.41	32.91	15.04	128.30
NET	-7.90	7.78	-8.65	6.23	2.54	
cTCI/TCI				32.08 / 25.66		

Notes: The table reports average connectedness measures from the TVP-VAR model using forecast horizon $H = 10$. Rows indicate the receivers, columns indicate transmitters. Diagonal elements report own-volatility shares, while off-diagonal elements report cross-variable volatility spillovers. All values are expressed in percentage points.